

Ludovic FLAMENT

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Date of birth: 6th February 1974

Married, two children

Cryptography, PKI, Common Criteria Expert - Owner

PROFESSIONAL BACKGROUND

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|---------------------------------|--|
| September 2022
– In progress | Consultant PKI (Cryptograph'IC / French minister) <ul style="list-style-type: none">• Migration of existing PKI and CMS to an external company Helping on review of specifications, risks analyses, process of outsourcing, use cases ... |
| September 2023
- In progress | Consultant Common Criteria evaluation (Cryptograph'IC) <p>Writing the Security Target for a file protection product (EAL4+ certification). Help developers for the cryptographic specifications and the documentation to provide to IT labs</p> |
| May 2021 -
January 2024 | Certification Coordinator (Cryptograph'IC / TheGreenBow) <ul style="list-style-type: none">• Certifications manager for CC, CSPN, FIPS certification of the VPN product• Writing and supervising the writing of technical documents and Security Target• Relation with CC Testing Labs and ANSSI (French government) |
| January 2023 –
April 2023 | Consultant PKI (Cryptograph'IC / French retail company) <ul style="list-style-type: none">• Technical audit of the PKI (usage, key generation, key activation, certificate profile, key storage, backup ...)• Organizational audit (policies, registry of request, validation process, revocation process ...)• Interviews with business unit owners to discuss the use of PKI in relation with actual and future security requirements• Recommendations on the technical and organizational measures to have a trusted PKI |
| April 2021 -
August 2021 | Consultant Common Criteria evaluation (Cryptograph'IC) <p>Writing the Security Target for a ground-to-air communication gateway product</p> |
| March 2020 -
December 2021 | Consultant Common Criteria evaluation (Cryptograph'IC) <p>Accompaniment for Common Criteria evaluation of a security product at level EAL4+</p> <ul style="list-style-type: none">• Writing and supervising the writing of technical documents• Relation with CC Testing Labs and ANSSI (French government) |
| January 2019 -
December 2019 | Consultant CSPN (Cryptograph'IC / Geoide) <ul style="list-style-type: none">• Helping to write Security Target and cryptographic specifications for a firewall editor for a First Level Security Certification (French CSPN) |
| January 2017 -
May 2021 | Consultant PKI (Cryptograph'IC / French minister) <ul style="list-style-type: none">• Migration of existing PKI, POC, drafting of technical (architecture, installation, operating, update ...) and organizational documents (Certificate Policy, Certification Practice Statement, Key Ceremony ...)• Highlighting of bugs in the product used, creation of corrective patches, relationship with the publisher for integration into future versions• Management of HSM (installation, configuration, update)• Dematerialization of certificate request process• Technical expertise for project using certificates• L3 Support |
| November 2018
– January 2020 | Consultant Common Criteria evaluation (Cryptograph'IC) <ul style="list-style-type: none">• Project Manager for a VPN client certification• Writing technical documents (functional, detailed and architecture specifications) |
| October 2018 –
February 2019 | Consultant cryptography (Cryptograph'IC) <ul style="list-style-type: none">• Design and development of a new protocol that permit the secure storage of information with timestamp and sequencing guarantee (similar to blockchain). |

July 2018 – February 2019	Consultant CSPN (Cryptograph'IC) Helping to write Security Target and cryptographic specifications for a firewall editor for a First Level Security Certification (French CSPN)
July 2017 – January 2018	Consultant cryptography (Cryptograph'IC / BCDiploma) Write a secure protocol to protect data information using Blockchain an compliant with RGD.
September 2016	Consultant CSPN (Cryptograph'IC) Helping to write Security Target for a network probe editor for a First Level Security Certification (French CSPN)
March 2016 – July 2016	Consultant cryptography (Cryptograph'IC / WolfSSL) Development of new cryptographic functions in wolfSSL cryptographic library: OpenSSL BIO like support, HSM support through PKCS#11.
December 2015 – February 2016	Consultant cryptography (Cryptograph'IC / Technicolor) - Development of tool that manage PKI using HSM to secure the keys - Development of cryptographic libraries for code signing and tickets for an SSO solution. Also using HSM to secure keys.
May 2015 – November 2015	Consultant cryptography (Cryptograph'IC / WolfSSL) Development of new cryptographic functions (X.509 extensions, ALPN, IDEA...) and integration in existing tools (OpenSSH) of wolfSSL (formerly CyaSSL) cryptographic library.
November 2014 – June 2015	Consultant cryptography / PKI (Cryptograph'IC / Imprimerie Nationale) - Consultant for a Card Management System on the PKI and Smartcard driver component - Project Manager for a Microsoft mini driver component
July 2014 – November 2014	Consultant cryptography (Cryptograph'IC / Energy ICT) - Development of a Java CSP, which delegates some cryptographic operations to a Secure Element Chip integrated on a Data Concentrator product, used to collect data securely from a large number of Smart Meters. - The connection between Java CSP and Secure Element is done by the integration of a PKCS#11 library.
January 2014 – March 2014	Consultant cryptography (Cryptograph'IC / Ingenico) Study on the development of a new service for an editor of security products
November 2013 – January 2014	Consultant Common Criteria evaluation (Cryptograph'IC / Morpho) Writing Security Target of a Smartcard for Common Criteria evaluation
October 2013	Consultant CSPN (Cryptograph'IC) Writing Security Target for an Antivirus editor for a First Level Security Certification (French CSPN)
July 2013 – October 2013	Consultant cryptography/PKI (Cryptograph'IC / GMX) - Design, development of service that deliver X.509 certificate for payment device - Service architecture is on client/server model with support of HSM
November 2012 – June 2013	Consultant cryptographic prototype (Cryptograph'IC / PMU) - Design, development of cryptographic API to secure the communications between end-user and central service - Development of a prototype for industrialization
March 2012 – June 2014	Consultant for accompaniment on Common Criteria evaluation (Cryptograph'IC / NETASQ) During an accompaniment for Common Criteria evaluation of a security product at level EAL3+/QS and EAL4+, these tasks will be accomplished: - Project management - Writing and Supervising the writing of technical documents - Relation and supervision of work done by CC Testing Labs - Relation with ANSSI (French government)

June 2011 - December 2012	Consultant for accompaniment on Common Criteria evaluation (Cryptograph'IC / Cryptolog) During an accompaniment for Common Criteria evaluation of a security product at level EAL3+/QS, these tasks will be accomplished: • Project management • Writing and Supervising the writing of technical documents • Relation with CC Testing Labs and ANSSI (French government)
March 2011 - March 2012	Consultant cryptographic API developer (Cryptograph'IC / ING Bank) • Design, development of cryptographic API • Encryption and signature with Hardware Security Module (with PKCS#11) • Client/Server version for no cryptographic capable OS like mainframe.
December 2010 - January 2011	Cryptographic expertise of a software (Cryptograph'IC / Orange) • Analyze source code • Test compliancy of algorithms with standard • Search and test attacks
January 2010 - December 2010	Consultant cryptographic API developer (Cryptograph'IC / ING Bank) • Design, development of cryptographic API • Encryption and signature with Hardware Security Module (with PKCS#11) • Client/Server version for no cryptographic capable OS like mainframe.
April 2010 - August 2011	Consultant for accompaniment on Common Criteria evaluation (Cryptograph'IC / ARKOON) During an accompaniment for Common Criteria evaluation of a security product at level EAL3+/QS, there tasks were accomplished: • Project management • Supervising the writing of technical documents • Relation with CC Testing Labs and ANSSI (French government)
November 2008 - December 2009	Consultant for accompaniment on Common Criteria evaluation / Cryptographic Expert / Team Leader (Cryptograph'IC / NETASQ) • Writing and Supervising the writing of technical documents • Relation with CC Testing Labs and DCSSI (French government) • Leader of 6 engineers • Supervision of projects (firmware release) • Report to direction • Manage and develop cryptographic evolution of products. This concern services for end user (VPN-SSL, authentication, ...) and products itself security (license, update, communication protocol, ...).
April 2008 – October 2008	Consultant for accompaniment on Common Criteria evaluation (Cryptograph'IC / NETASQ) During an accompaniment for Common Criteria evaluation of a security product (Level EAL3+/QS and EAL4+), there tasks were accomplished: • Help for definition of perimeter • Writing of Security Target and technical documents • Supervision technical documents writing • Relation with CC Testing Labs and DCSSI (French government)
February 2008	Cryptographie Ingénierie & Conseil (Cryptograph'IC) WebSite : http://www.cryptograph-ic.com Founder and CEO of an independent company specialized on cryptography and PKI services professional. ^[1] This company will mainly deliver expertise, development, architecture, protocol design, ... for company that will be use cryptography and PKI.
From June 2007 to February 2008	Cryptographic Consultant: SONY NSCE company Working on Marlin DRM technology for TV products and mainly developed a technological demonstration product.

- From June 2001 to June 2007 **Project manager, Cryptographic Engineer: *NETASQ company***, French unified security appliances manufacturer (<http://www.netasq.com>).
Position (from February 2006 to June 2007): Project/Team manager
- Management of engineer team (4 people)
 - Supervision of projects (firmware release)
- Position: Cryptography and PKI expert
- Manage and develop cryptographic evolution of products. This concern services for end user (VPN-SSL, authentication, ...) and products itself security (license, update, communication protocol, ...).
 - Manage and develop PKI service
 - Participate on the Common Criteria evaluation
- From October 1999 to June 2001 **Research and development Engineer: *CERTPLUS company (French certification operator)***
- development of new service (mainly a key recovery mechanism with smartcard support) for the PKI platform of company
 - Tier 5 Verisign for the customer's PKI
- From 1999 to 2010 **Freelance developer of cryptographic library:**
Authorized by the French government for commercial purposes
- Big number computing on Galois Field $GF(p)$ / p prime
 - Polynomial computing on Galois Field $GF(2^n)$
 - Secret sharing: Shamir algorithm.
 - Compression before ciphering with ZLIB library.
 - Pseudo Random Number Generation: algorithm FIPS186-2, ANSI X9.17 & Mersenne Twister.
 - Asymmetric algorithm that use Elliptic Curve (standard IEEE P1363-2000 et P1363a)
 - Symmetric algorithm.
 - Hash, MAC & HMAC functions.

EDUCATIONAL BACKGROUND

- June 1999 **Master's Degree in computer science** (University for Science and Technology of Lille - FRANCE)
- Software for polynomials factorization, Berlekamp algorithm.
 - Software for digital signature, Nyberg-Rueppel elliptic curve algorithm.
- June 1998 **Degree in computer science** (Jean-Perrin Faculty, Lens - FRANCE)
- Software for compression of data, LZW algorithm.
 - Software for memory allocation, reallocation, free, ...

Computer Science:

- Programming language: C, C++, Java.
- Interpreted language: Shell UNIX, PERL, HTML.
- Operating system: FreeBSD, MAC OS-X, Windows, Linux, Sun-Solaris.
- Cryptography and security:
 - *Cryptographic token:* smartcard, USB token, SSL accelerators card, Hardware Security Modules, ...
 - *Standards and Protocols:* PKCS, X509, SSL, IEEE P1363-2000, ANSI X9, PKI, SRP, RFC on cryptographic algorithms, ...
 - *Miscellaneous:* very good knowledge of OpenSSL and cryptographic algorithms: symmetric, asymmetric, hash function, authentication, ...

Professional activities:

- Conference on SRP protocol, PKI, VPN during meeting/show on security area.
- Article on SRP protocol for the French MISC magazine number 15 (September/October 2004)
- Article on elliptic curve for the French MISC magazine number 19 (May/June 2005)
- Interview on the French magazine MAG-SECURS, number 21 (October/December 2008)

Foreign Languages:

- Fluent in French (mother tongue)
- Working knowledge of English

OTHER ACTIVITIES

Driving license, Personal car
 Sports: Fitness, Running, Chess (International ELO), Walk in nature
 Hobbies: Photography, Sciences, Economy.